

TIRTH PATEL

AI/ML & Data Science

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SUMMARY

B.Tech Computer Engineering student specializing in AI/ML, data science, Python, SQL, and data analysis. Experienced in data cleaning, EDA, statistical analysis, and supervised learning with Scikit-learn. Built Power BI dashboards for KPI tracking. Eager to contribute to AI/ML and Data Science teams by turning raw data into actionable insights.

TECHNICAL SKILLS

Data Analysis: Data Cleaning, Data Wrangling, EDA, Statistical Analysis, Pandas, NumPy, Excel (Pivot Tables, VLOOKUP, Charts)
Machine Learning: Supervised Learning, Regression, Classification, Scikit-learn, Model Evaluation, Feature Engineering, XGBoost
AI & Computer Vision: YOLOv8, OpenCV, Generative AI, LLM APIs (Groq), LangChain, Geospatial Data (Rasterio)
Visualization: Matplotlib, Seaborn, Power BI, KPI Dashboards, Business Intelligence, Leaflet (GIS Maps)
Programming & Databases: Python, SQL, MySQL, Relational Databases, SQL Joins, Subqueries, Aggregations
Deployment & Tools: FastAPI, REST APIs, Docker, Vercel, Hugging Face, Git/GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECT EXPERIENCE

NER AI Landslide Early Warning System — Python | FastAPI | Machine Learning | YOLOv8 | GenAI [GitHub](#)

- Built an AI early-warning platform predicting landslide risk across 8 North Eastern Indian states from rainfall, soil moisture, terrain, and land-cover features using ML; served via FastAPI with an interactive Leaflet GIS map.
- Integrated YOLOv8 hazard detection on field images, live weather data, Groq LLM multilingual alerts, and automated Email/SMS (Twilio) alerts for HIGH/CRITICAL risk; deployed on Vercel with models on Hugging Face.

AQI Analysis — Python | Pandas | Matplotlib | Seaborn [GitHub](#)

- Analysed 9,000+ air quality records (2004–2005) to identify pollution trends, seasonal AQI patterns, and key pollutant correlations; built interactive Matplotlib/Seaborn visualizations for environmental monitoring.

India Economy Analysis — Python | EDA | Statistical Analysis [GitHub](#)

- Performed EDA and statistical analysis on 12 years of economic data (2011–2023) covering GDP growth, inflation, and sector-wise trends; developed KPI dashboards for business and policy decision-making.

Diabetes Trend Analysis — Excel | Pivot Tables | Dashboards [GitHub](#)

- Cleaned and analysed a healthcare dataset of 250,000+ patient records to identify diabetes risk patterns; built Pivot Table dashboards and charts for stakeholder health-insight reporting.

Sales Analysis Dashboard — Power BI | KPI Tracking | BI [GitHub](#)

- Built an end-to-end interactive Power BI dashboard tracking 10+ KPIs across product categories, regions, and time periods, applying data cleaning and wrangling for stakeholder decision-making.

Crop Recommendation Dashboard — Power BI | Agricultural Data [GitHub](#)

- Developed a data-driven BI dashboard recommending optimal crops based on environmental and soil factors, visualizing key variables to support data-driven farming decisions.

Hospital Management System — SQL | MySQL | Relational Databases [GitHub](#)

- Extracted, aggregated, and analysed relational hospital data across 5+ linked tables using advanced SQL joins, subqueries, and aggregations to solve operational business problems.

World Happiness Analysis — Python | Jupyter | EDA | Data Visualization [GitHub](#)

- Performed EDA on the World Happiness Report to analyse how GDP, social support, life expectancy, freedom, and corruption drive happiness scores.

EDUCATION

B.Tech in Computer Science & Engineering (2024 – 2028, Expected)

Gandhinagar Institute Of Technology, Gandhinagar, Gujarat

Higher Secondary Certificate – Science (2024)

Narayan Vidhyalay

CERTIFICATIONS & ACHIEVEMENTS

Advanced Training in Data Science & AI/ML — Red & White Skill Education, Ahmedabad | Jul 2025 – Sep 2026 | 20+ hands-on projects: Python, SQL, EDA, Statistical Analysis, Power BI, Supervised ML

Certificate of Training Completion – Data Science — Acmegrade & IIT Delhi Rendezvous | Sep 2025 – Oct 2025